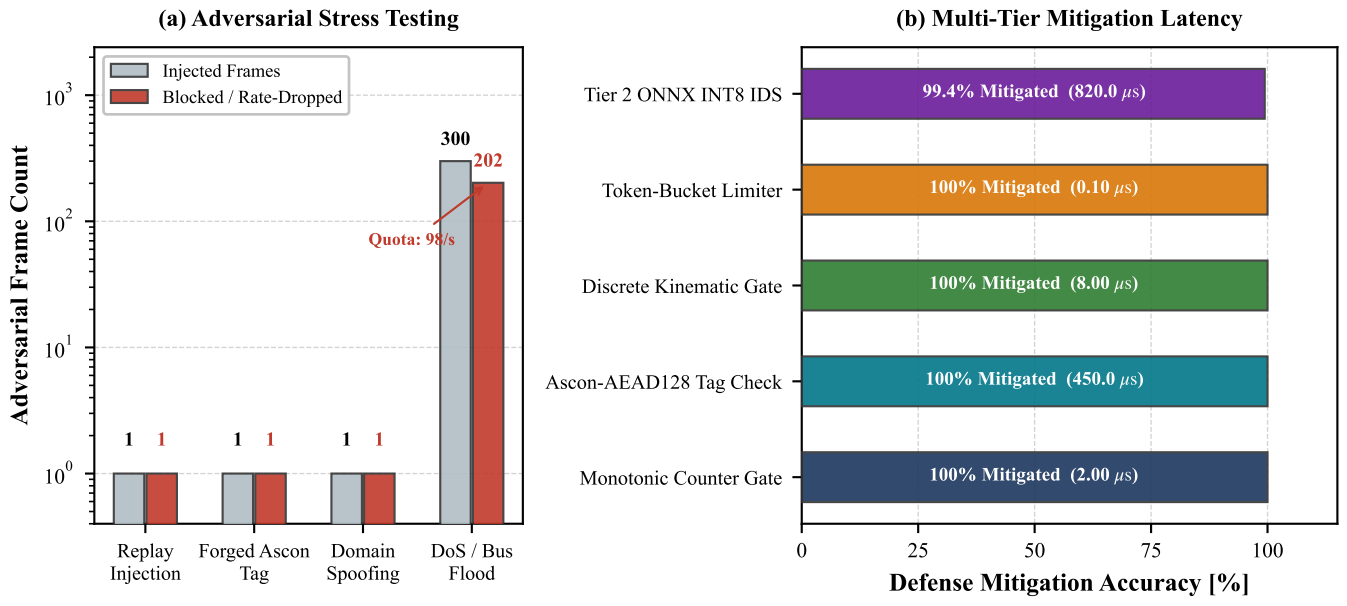
(a) Adversarial Stress Testing
Injected Frames
Blocked / Rate-Dropped
Adversarial Frame Count
10^3
10^2
10^1
10^0
300
202
Quota: 98/s
1
1
1
1
1
1
Replay Injection
Forged Ascon Tag
Domain Spoofing
DoS / Bus Flood
(b) Multi-Tier Mitigation Latency
Tier 2 ONNX INT8 IDS
99.4% Mitigated (820.0 μs)
Token-Bucket Limiter
100% Mitigated (0.10 μs)
Discrete Kinematic Gate
100% Mitigated (8.00 μs)
Ascon-AEAD128 Tag Check
100% Mitigated (450.0 μs)
Monotonic Counter Gate
100% Mitigated (2.00 μs)
0
25
50
75
100
Defense Mitigation Accuracy [%]